

Unsupervised Functional Stage Detection From EEG Using Topological Data Analysis

Aleksandr Abramov, Ekaterina Mikhaylets, Vsevolod Chernyshev, Alexander Kaplan, Fedor Ratnikov

HSE University (Moscow) Ulm University Lomonosov Moscow State University

Objectives

Assess whether **topological data analysis (TDA)** can support unsupervised detection of functional EEG stages with the state-detecting algorithm (SDA) [1]:

- Test TDA as an alternative to traditional spectral features for stage boundary detection
- Compare both approaches on Guhyasamaja meditation EEG and public sleep-edf recordings
- Determine whether topology complements or replaces spectral structure in this task

Introduction

EEG segmentation is largely manual. Recently presented **SDA** [1] solves the task without knowing event count or timing, and works well with spectral features (PSD, PLV, coherence). These descriptors capture frequency and correlation structure but not the spatial geometry of multichannel recordings.

TDA studies shape and connectivity via persistent homology [2, 3]. We assess whether this topological view reveals functional stages that spectral methods miss.

TDA Feature Engineering

Fig. 1 summarizes the three-branch topological extraction pipeline applied to each EEG epoch. Fig. 2 shows how filtered persistence diagrams are vectorized into machine-learning-ready descriptors.

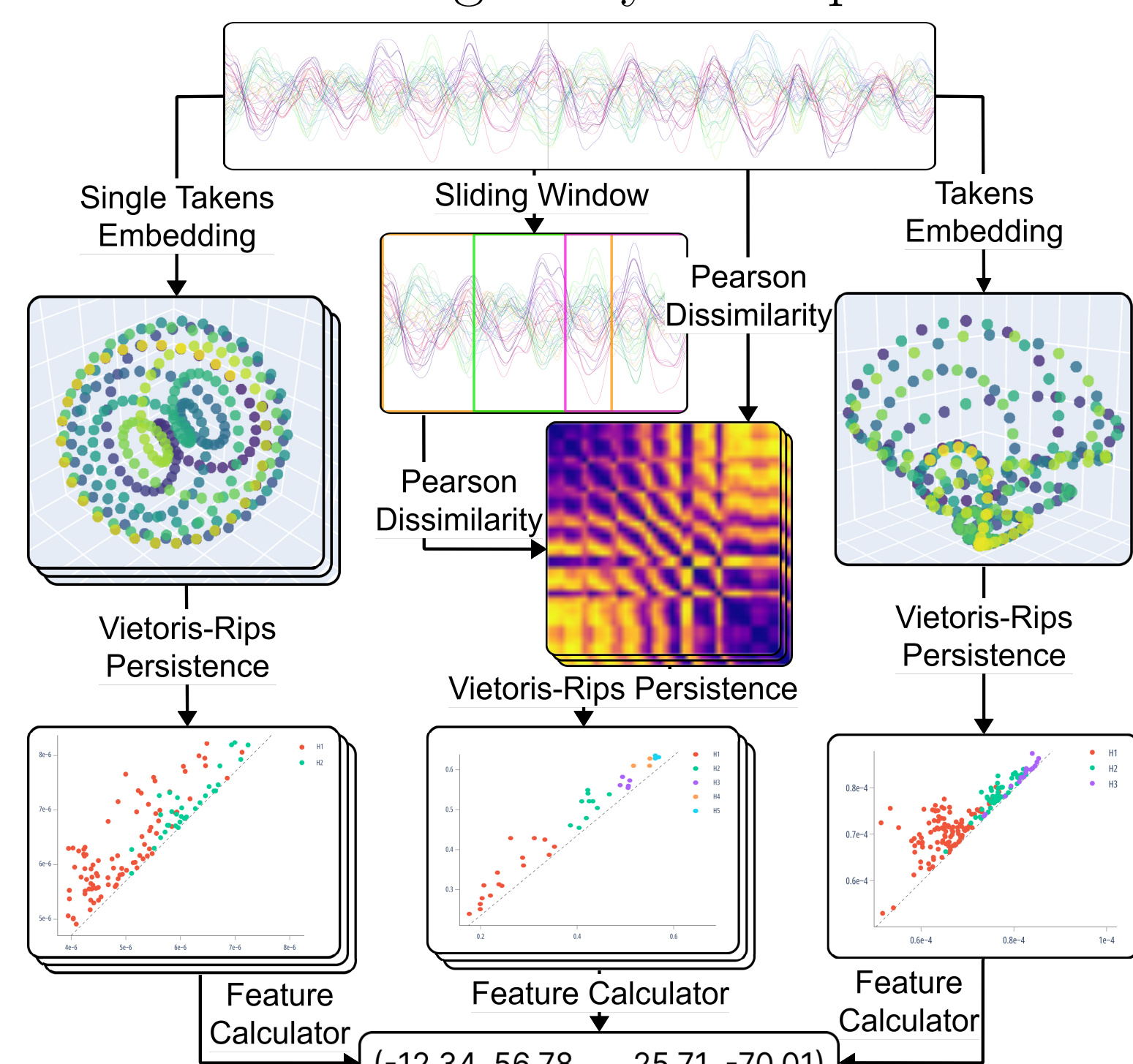


Fig. 1: Topological feature extraction pipeline.

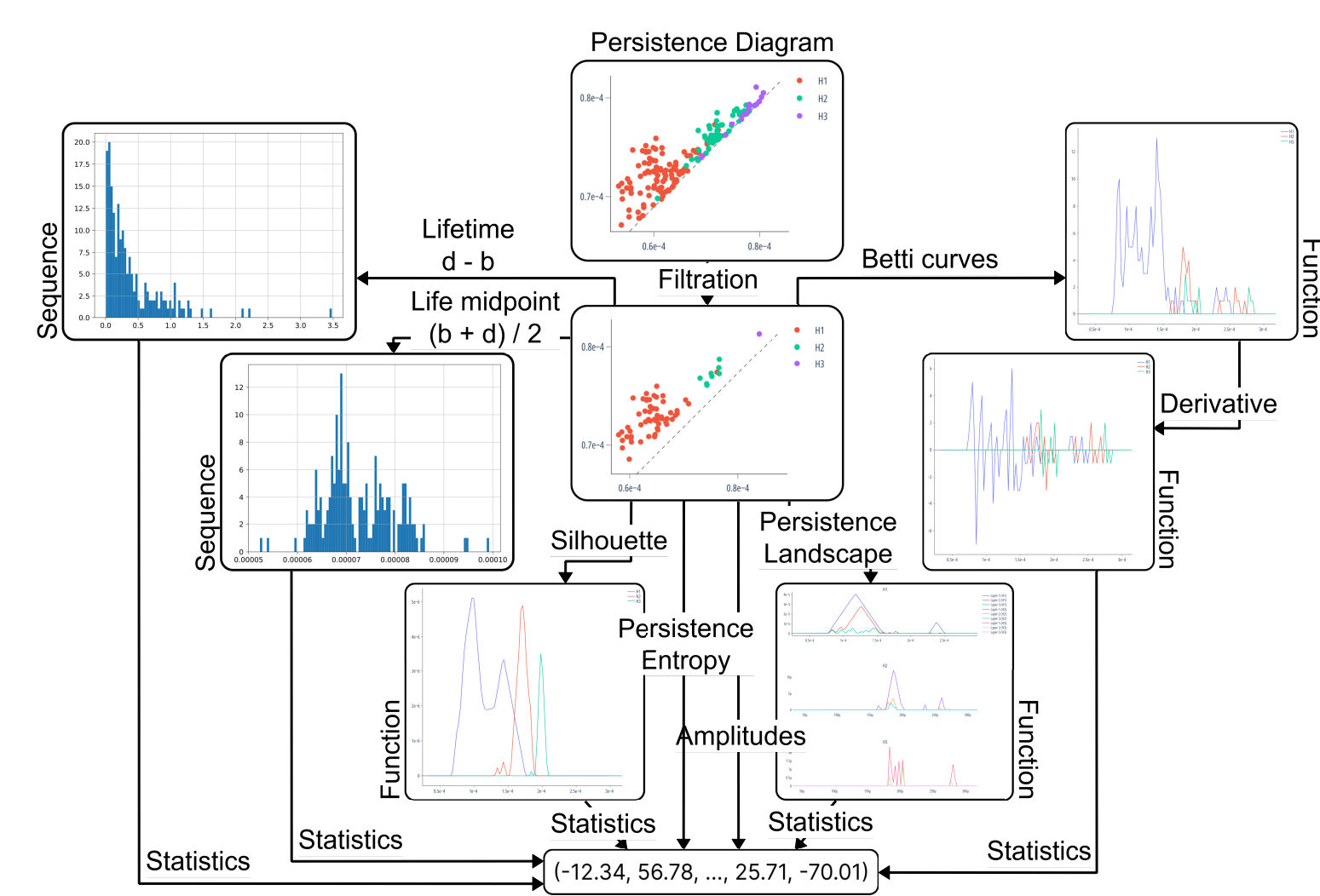


Fig. 2: Persistence diagram vectorization.

Datasets

Meditation.

- 30 Tibetan Buddhist monks practicing Guhyasamaja Tantra; 8 canonical stages; recordings from three field expeditions
- 38 or 32 scalp electrodes split into 11 regions of interest; 1-s epochs after artifact removal; compared against 765 spectral baseline features from [1]

Sleep.

- 153 whole-night PSG recordings from sleep-edf [4] (Sleep Cassette Study)
- Resampled to 50 Hz; 15-s non-overlapping epochs; expert stage annotations for external evaluation
- At most 10 functional stages predicted per subject, matched to expert labels

Methods

- **SDA** [1]: Ward clustering with temporal connectivity; K-Means consensus; **adapted silhouette** quality metric
- **TDA**: three extraction branches (pipeline in Figs. 1–2) yield $\sim 20,000$ descriptors: Taken's embeddings & Pearson dissimilarity $\rightarrow$ Vietoris–Rips filtrations $\rightarrow$ persistence diagrams $\rightarrow$ vectorization $\rightarrow$ 15 PCA components.
- **QSDA & IV**: feature selection based on single-feature SDA ranking and information-value filtering of weak descriptors
- **Experiments**: spectral vs. topology on meditation; surrogate data and statistical tests for validation; three feature sets on sleep vs. expert labels

Validation

Robustness. Noise changed scores by at most a factor of three; shuffling collapsed quality by tens of times. Best boundaries emerged from combined descriptors. Mann–Whitney indicates 30% of features differ significantly between adjacent stages (none for shuffled epochs).

Feature quality. QSDA and IV agree on valuable descriptors; the space was pruned from $\sim 20,000$ to $\sim 4,500$ features. Frontal electrodes dominate on average, with subject-specific variation. Amplitude means, vector norms, and persistence silhouettes ranked highest; percentiles and skew were least informative. Pearson dissimilarity features exposed interesting patterns but ranked lowest overall and were pruned.

Conclusion

TDA is a practical robust alternative to spectral features for unsupervised SDA-based EEG stage detection on continuous processes. Topology and spectra capture **complementary** structure and are best used together. **QSDA** efficiently prunes unimportant descriptors and scales well. Exploring representative cocycles is key for interpretability and future work.

References

- [1] Ekaterina Mikhaylets, Alexandra M. Razorenova, Vsevolod Chernyshev, et al. Sda: a data-driven algorithm that detects functional states applied to the eeg of guhyasamaja meditation. *Frontiers in Neuroinformatics*, 17, 2024.
- [2] H. Edelsbrunner and J. Harer. *Computational Topology: An Introduction*. Applied Mathematics. American Mathematical Society, 2010.
- [3] Facundo Mémoli and Kritika Singhal. A primer on persistent homology of finite metric spaces. *Bulletin of Mathematical Biology*, 81(7):2074–2116, May 2019.
- [4] B. Kemp, A.H. Zwinderman, B. Tuk, H.A.C. Kamphuisen, and J.J.L. Obery. Analysis of a sleep-dependent neuronal feedback loop: the slow-wave microcontinuity of the eeg. *IEEE Transactions on Biomedical Engineering*, 47(9):1185–1194, 2000.

Acknowledgements

Supported by the Russian Science Foundation, Grant No. 24-68-00030.

Contact

- asabramov@edu.hse.ru
- Faculty of Computer Science, HSE University

Important Result

- Topological features **match or outperform** spectral baselines for SDA on both meditation and sleep-edf datasets, and are more consistent with expert labels
- **QSDA** improves over the full descriptor set and is coherent with IV; frontal electrodes carry the most informative features on average (Figs. 3, 5)
- TDA captures **complementary geometry** missed by spectra — including high-dimensional cocycles at meditation transitions (Fig. 4)
- Quality gains **justify** the computational expenses with potential for optimization
- **Statistical significance** is confirmed with surrogate data, extensive experiments, and Mann–Whitney U-test with Bonferroni correction

Results

Meditation. QSDA-filtered topological features beat the 765 spectral baseline for most subjects (Fig. 3) yet agree on the most stable boundaries. Persistence amplitudes from frontal electrodes rank highest on average (Fig. 5); Subject 2 cocycles (Fig. 4) localize to limbic regions at stages 1, 4, and 7, highlighting entry to and exit from meditation. Combined descriptors often yield the most compact clusters.

Sleep. Topology yields denser, more expert-aligned partitions with consistently tighter scatter-plots; combined spectral and topology features add no benefit. With seven PSG channels and no QSDA pruning, unchanged hyperparameters still favored topology. Ten sought stages matched interpretable expert-label subsets: adapted silhouette $0.030 \pm 0.009 \rightarrow 0.228 \pm 0.010$; Fowlkes–Mallows index $0.843 \pm 0.017 \rightarrow 0.911 \pm 0.013$.

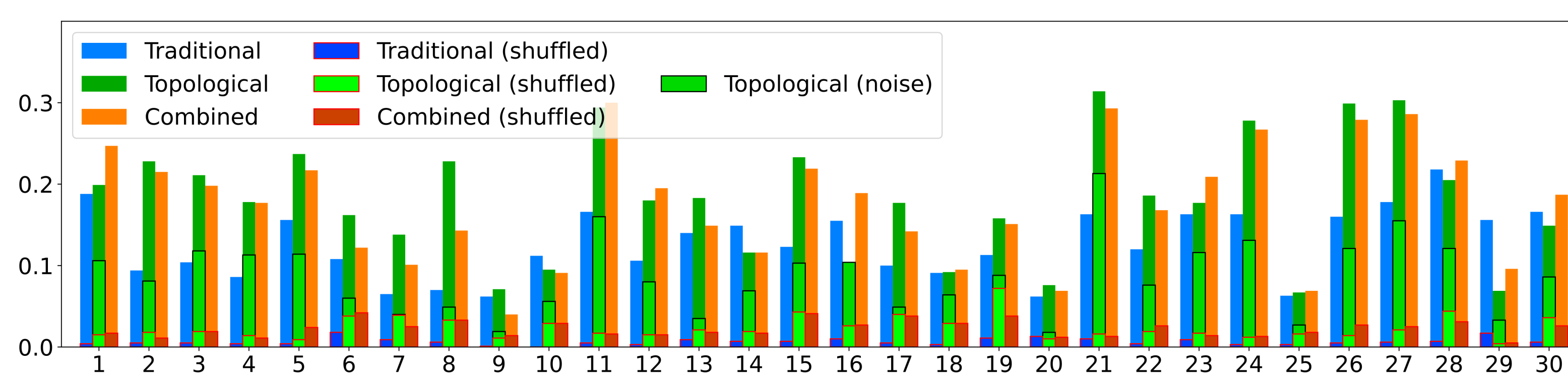


Fig. 3: Adapted silhouette for 30 meditation subjects (original vs. shuffled).

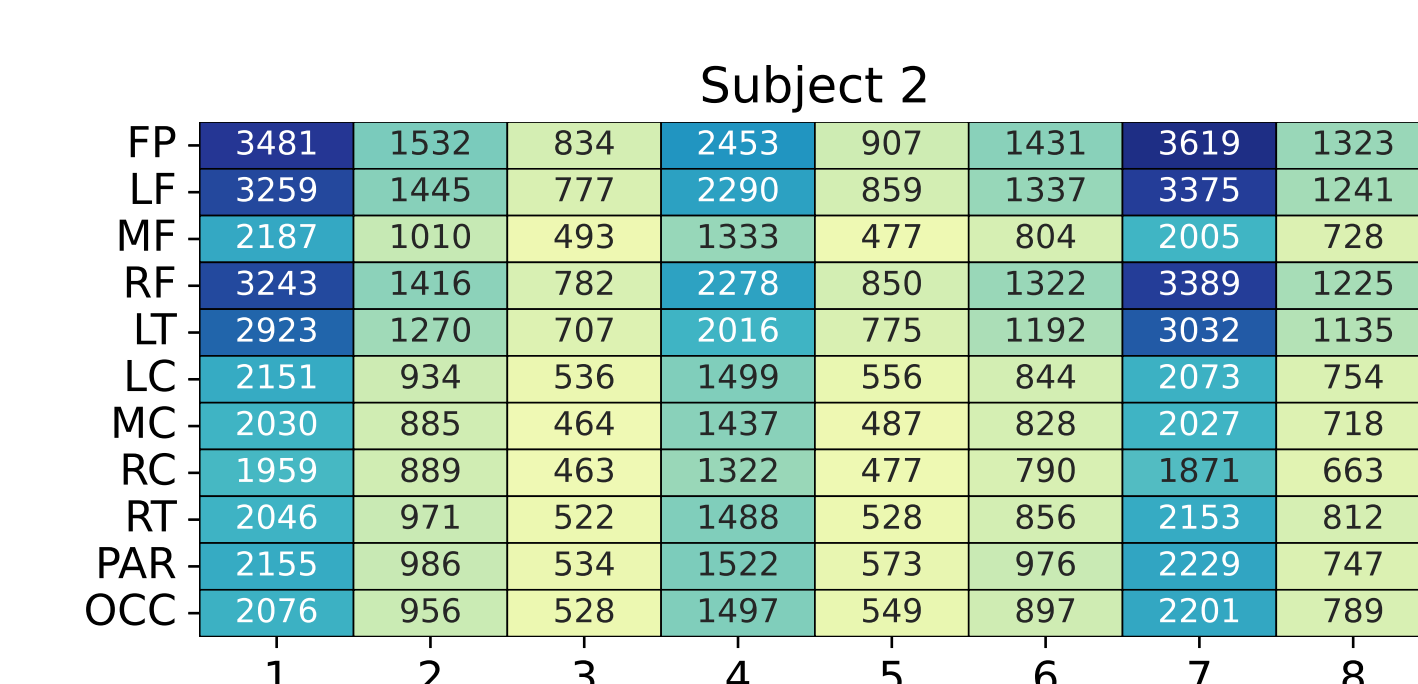


Fig. 4: Dimension-5 representative cocycles by brain region and stage (Subject 2).

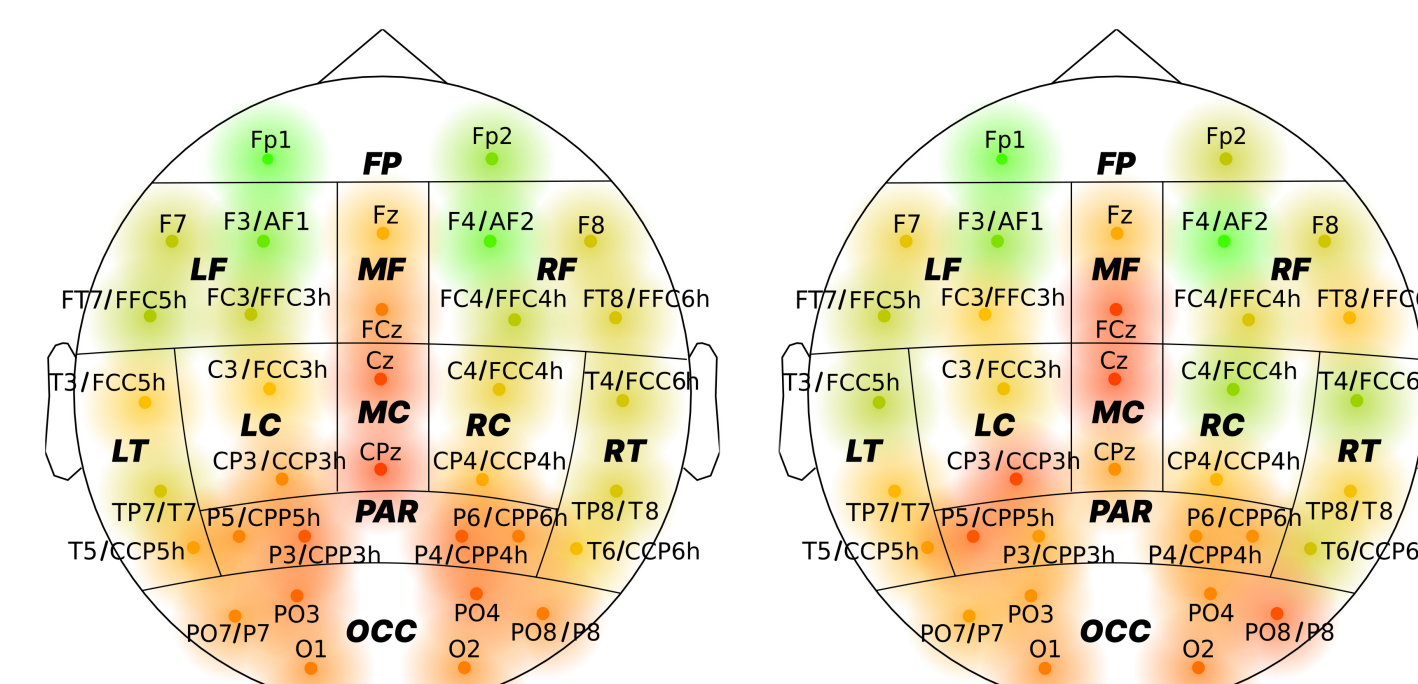


Fig. 5: QSDA scores by electrode (30 subjects).